Mapping Material Responses and Failure Mechanisms for a Heat-Sink Rotating Detonation Engine

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Introduction

Rotating Detonation Engines (RDEs) have seen rapid advancement in recent years, yet the extreme thermomechanical environment they impose on the walls of the combustion chamber remains a critical barrier to practical implementation. Existing work has largely focused on quantifying wall heat flux and temperature [1–5] and on system operability [6–9], with little attention given to the underlying mechanisms of component failure.

Experimental Platform and Copper Baseline

To investigate the failure mechanisms directly, we developed a methane-oxygen RDE as a material testing platform with an outer annulus diameter of 2 in and a removable copper inner body that serves as the test article [10]. Test conditions of a total propellant mass flow rate of 200 g/s and $\phi = 1.15$ were selected to represent in-space, uncooled RDE operation [11], where burns are short and components cool fully between firings. At these test conditions, we obtained two-wave operation with a wave frequency of 18 kHz and wave speed of 1450 m/s. Accordingly, the engine was fired in 2 s intervals, cooling below 50 °C (measured using an optical pyrometer) between each firing, for a cumulative 120 s of runtime. Every 20 s, the inner body was removed for characterization along its full axial length. Diameter measurements, taken at micron resolution, tracked bulk dimensional change over the cumulative runtime. Digital microscopy provided qualitative surface imaging to identify and track evolving damage features, while laser confocal microscopy quantified surface height changes with high spatial resolution — together capturing both the morphological and topographical evolution of the surface.

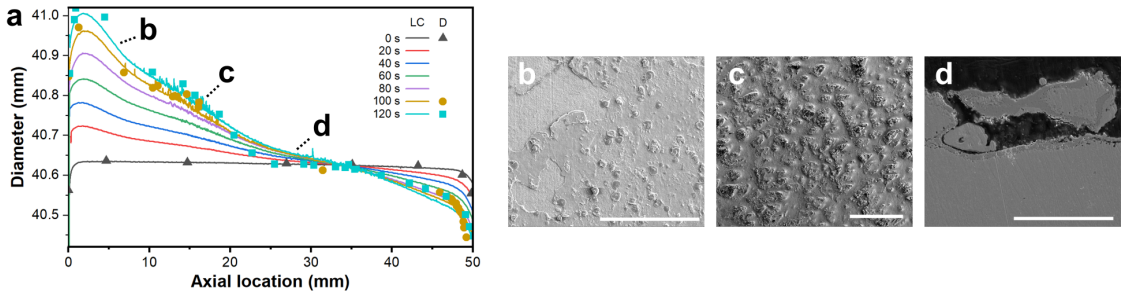


Fig. 1 a) Inner body diameter profiles measured by laser confocal (LC) and diameter (D) methods at each characterization interval over the cumulative 120 s **runtime; x = 0 mm is the chamber inlet. b) SEM image near the injector face; scale bar** 300 µm**. c) SEM image near the peak detonation height; scale bar** 50 µm**. d) Cross-sectional SEM image at the axial midpoint of the inner body; scale bar** 30 µm**.**

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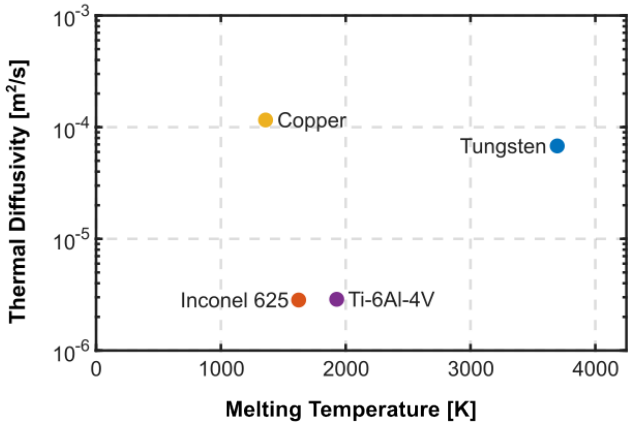


Fig. 2 Thermal diffusivity vs. melting temperature for the four candidate materials, spanning distinct regions of the property space.

The confocal microscope images coupled with the diameter measurements allowed us to track the total diameter to micron resolution across the entire axial distance shown in Fig. 1a. The surface diameter profile evolved dramatically, with the injector side growing nearly 400 µm in diameter. The combustor exit end shrank roughly 100 µm with an additional rapid drop-off at the edges of the test article. While some have proposed that crack formation due to high cycle fatigue from the detonation pressure wave may be a failure mechanism [12], none was observed over the 2.1 million detonation pressure cycles accumulated over 120 s of runtime. Rather, scanning electron microscopy (SEM) imaging of the inner body revealed distinct regions of damage at distinct axial locations. Very close to the injector face (Fig. 1b), there were regions of large flat oxide layers with areas of pure copper exposed, where polishing marks from surface preparation were still visible on the bare copper. Further down the body, coinciding with the detonation height in the chamber (Fig. 1c), blobs roughly 100 µm across dotted the surface; chemical composition maps revealed these to be carbon and zinc, the latter likely from vaporization of the brass injector during engine operation. A cross-sectional SEM image (Fig. 1d) revealed a blob of copper completely disconnected from the surface, indicating that the surface softened to near its melting point and shed material from the body. The lack of crack formation points toward a failure mechanism governed by ablation and melting, which is in turn dictated by the material's global thermal properties.

Candidate Material Selection

Building on the copper baseline, we extend the platform to benchmark candidate replacement materials, using bulk thermal properties as the basis for comparison. We selected four materials spanning distinct regions of thermal diffusivity vs. T_{melt} (melting temperature) space — Copper, Inconel 625, Tungsten, and Ti-6Al-4V, plotted in Fig. 2 — chosen both to cover the relevant property space and to represent classes already used in propulsion hardware. Inconel 625 is a baseline nickel superalloy widely employed in combustion chamber liners and hot-section components; Ti-6Al-4V is a workhorse structural alloy used throughout aerospace propulsion systems; tungsten anchors the refractory extreme and is representative of throat and high-flux insert materials; and Copper both continues the present baseline study and serves as a high-conductivity reference relevant to regeneratively cooled chamber walls. We expect the two high-diffusivity materials (Copper and Tungsten) to show a smaller surface-temperature rise than the two low-diffusivity materials (Inconel 625 and Ti-6Al-4V).

As a preliminary analysis, we ran 1D thermal diffusion simulations modeling the inner body as a solid cylinder, as shown in Fig. 3a. Boundary a_1 is treated as adiabatic, while a_2, the inner-body surface exposed to the RDE chamber, is assigned a Robin boundary condition with T_{gas} fixed at 1700 K and h set to 10 000 W/(m^2 K), consistent with methane-oxygen RDE estimates [3]. This analysis is intended only for relative comparison between materials: the cyclic, azimuthally non-uniform detonation heating is approximated as a steady, uniform convective load; oxidation, ablation, and radiation are neglected; and h is held constant, whereas in practice it varies with axial location and detonation phase. The reported temperatures should therefore be interpreted as relative rankings only, not as quantitative predictions of in-engine surface conditions. Figure 3b shows the temperature profiles for each material after 2 s of heating. Surface temperatures at $t = 2$ s rank in reverse order of diffusivity. Because we are concerned with surface melting and erosion,

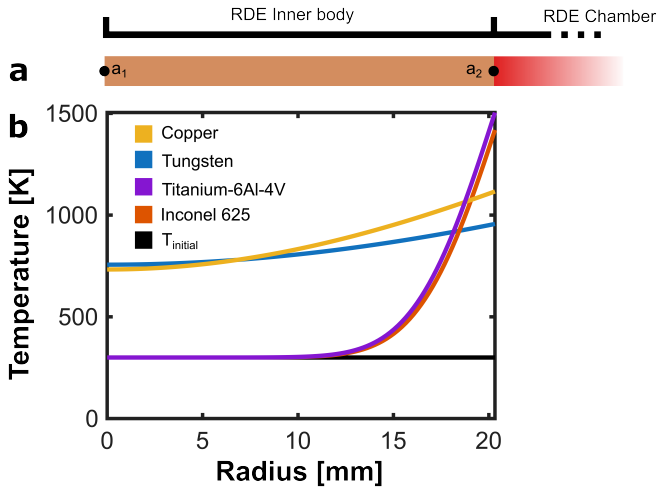


Fig. 3 One-dimensional thermal diffusion model used to rank candidate wall materials. a) Cylindrical model setup: boundary a_1 is treated as adiabatic, while boundary a_2 (inner-body surface exposed to the RDE chamber) is assigned a Robin (convective) boundary condition with prescribed gas temperature $T_{\mathrm{gas}} = 1700\,\mathrm{K}$ and convective heat-transfer coefficient $h = 10\,000\,\mathrm{W/(m^2\,K)}$. b) Through-thickness temperature profiles in each candidate material after 2 s of heating, corresponding to a single firing duration. The model is intended for relative material comparison only and neglects cyclic heating, oxidation, ablation, and radiation.

the more relevant metric is the homologous surface temperature, defined as peak surface temperature normalized by melting point. Table 1 summarizes these values. Notably, while Ti-6Al-4V reaches a T_{max} roughly 100 K higher than Inconel 625, Inconel 625 is significantly closer to its melting temperature — suggesting that Inconel 625 may be more susceptible to surface softening and ablation despite its lower absolute peak temperature.

Material	$T_{\mathrm{max\ surface}}$ [K]	T_{melt} [K]	$T_{\mathrm{max\ surface}}/T_{\mathrm{melt}}$
Copper	956	1358	0.70
Inconel 625	1418	1563	0.91
Tungsten	1116	3695	0.30
Ti-6Al-4V	1504	1877	0.80

Table 1 Peak surface temperature and melting point comparison for candidate RDE wall materials under 1D thermal analysis.

Testing Plan

These simulations provide a ranked prediction of surface thermal loading across the four materials, under the assumption that melting and ablation remain the dominant failure mechanism. Each material will be tested in the RDE platform under conditions identical to the copper case, with the same suite of characterization methods applied to track surface evolution. Each candidate is expected to operate in a distinct oxidation regime at the surface temperatures predicted by the 1D model. Inconel 625 forms a protective Cr_2O_3-based scale with slow, parabolic growth, which can mitigate surface recession but may spall under repeated thermal cycling. Ti-6Al-4V develops a mixed TiO_2/Al_2O_3 scale that is generally protective at moderate temperatures but becomes brittle and oxygen-permeable above roughly 600 °C, opening pathways for sub-surface embrittlement. Tungsten, despite its high melting point, undergoes catastrophic, non-protective oxidation above approximately 600 °C through formation of volatile WO_3 species, and is therefore expected to be limited by oxidative recession rather than melting in this environment. Because these mechanisms scale very differently with temperature, the failure mode predicted by the present thermal model may not match that observed in test — a question the experimental campaign is designed to directly answer. The final paper will report the

corresponding surface evolution, dimensional change, and post-test microscopy for all four candidate materials tested under identical operating conditions, with the goal of identifying the dominant failure mechanism in each and assessing how well a purely thermal ranking predicts the observed behavior.

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